

Why RMSNorm is faster: one fewer reduction stage

“Two-pass” and “single-pass” name the reduction stages; what costs bandwidth is the number of input traversals.

RMSNorm

read $x \rightarrow \Sigma x^2$

read $x \rightarrow$ normalise $\gamma \rightarrow$ write y

2 traversals $\cdot 2R+1W$

LayerNorm

read $x \rightarrow$ mean

read $x \rightarrow$ variance (centred)

read $x \rightarrow$ normalise $\gamma \cdot \beta \rightarrow$ write y

3 traversals $\cdot 3R+1W$

traffic



$R \ R \ W = 3$ units



$R \ R \ R \ W = 4$ units

$4 \div 3 = 1.33\times$ more traffic per element for LayerNorm — the structural source of the gap.